

Appendix A. Broader Applications and Extensions

Our stochastic differential equation framework extends naturally beyond code generation to diverse multi-objective LLM scenarios. In content generation systems balancing creativity, factual accuracy, and engagement

($\mathbf{x} = [\textit{creativity}, \textit{accuracy}, \textit{engagement}]^T$), the interference matrix reveals systematic trade-offs between creative expression and factual precision, enabling principled design of content strategies. For reasoning and decision support optimizing speed, thoroughness, and interpretability ($\mathbf{x} = [\textit{response_time}^{-1}, \textit{completeness}, \textit{explainability}]^T$), eigenvalue analysis identifies whether rapid responses necessarily compromise thoroughness or if balanced approaches exist. In human-AI collaboration systems balancing autonomy, user control, and task efficiency

($\mathbf{x} = [\textit{automation_level}, \textit{user_agency}, \textit{task_completion}]^T$), the framework enables analysis of collaboration dynamics and optimal handoff strategies. Multi-modal integration systems combining text, vision, and audio with objectives like accuracy, latency, and resource usage ($\mathbf{x} = [\textit{multimodal_accuracy}, \textit{response_latency}^{-1}, \textit{computational_efficiency}]^T$) benefit from our approach revealing how modal integration affects objective trade-offs and guides architecture design. For safety-critical applications balancing helpfulness, harmlessness, and honesty

($\mathbf{x} = [\textit{helpfulness}, \textit{safety}, \textit{truthfulness}]^T$), interference matrix analysis quantifies fundamental tensions in AI alignment and informs safety protocols. Each application requires domain-specific drift function modeling $\boldsymbol{\mu}(\mathbf{x}, \pi)$ and diffusion characterization $\boldsymbol{\sigma}(\mathbf{x}, \pi)$, but the underlying mathematical framework remains universal, positioning our approach as a foundational tool for multi-objective LLM system design across diverse domains.

Appendix B. Dynamical Systems Insights and Algorithm Design

Our framework provides principled guidelines for designing interaction strategies based on desired dynamical properties:-

B.1. Convergence Rate Control

The eigenvalue spectrum directly controls convergence behavior. For rapid stabilization, choose strategies yielding real negative eigenvalues with large magnitude. For applications requiring exploration of multiple solutions, complex eigenvalues provide controlled oscillatory search.

B.2. Predictability vs. Performance Trade-offs

Balanced strategies ($\boldsymbol{\mu}$ with uniform coefficients) achieve higher predictability ($R^2 = 0.74$) but may converge to suboptimal equilibria. Focused strategies sacrifice predictability for potential performance gains in specific objectives.

B.3. Interference-Aware Strategy Design

The interference matrix guides strategy selection. When strong negative correlations exist (e.g., functionality vs. efficiency: $I_{fe} = -0.17$), sequential optimization (optimizing one

objective first, then others) may outperform simultaneous approaches. These interference patterns generate testable hypotheses about strategy effectiveness: for instance, whether addressing strongly coupled objectives sequentially rather than simultaneously might improve overall performance.

B.4. Adaptive Strategy Switching

Dynamic strategy adaptation based on current position in objective space:-

1. **Exploration Phase:** Use strategies with complex eigenvalues for broad objective space coverage.
2. **Exploitation Phase:** Switch to strategies with real negative eigenvalues for rapid convergence.
3. **Boundary Avoidance:** Monitor eigenvalue proximity to zero and switch before extreme trade-offs.

B.5. Multi-Objective Algorithm Framework

Our results suggest a general algorithmic framework:-

1. **Initialize:** Choose balanced strategy for stable baseline,
2. **Analyze:** Compute local interference matrix and eigenvalue spectrum,
3. **Adapt:** Select strategy based on desired dynamical properties,
4. **Monitor:** Track convergence indicators and boundary proximity, and
5. **Switch:** Dynamically adjust strategy to maintain desired trajectory.

This framework transforms multi-objective LLM optimization from ad-hoc prompting to principled dynamical systems control.

Appendix C. Optimal Prompting Strategies

C.1. Empirically-Informed Strategy Selection

Based on convergence analysis, we propose adaptive strategies:-

First, use FF strategies for 2-3 iterations to establish functionality baseline, accepting security degradation. **Then**, switch to SF strategies for 3-4 iterations to address vulnerability accumulation. **Later**, apply EF strategies for final 2-3 iterations to optimize performance. Finally, use AI strategies throughout for maintenance and balanced improvements.

C.2. Intervention Triggers

Human intervention should occur when: Security scores drop below 2.0 (indicating critical vulnerabilities), Efficiency degrades by $> 30\%$ between iterations, and/or Convergence rate exceeds 1.5 (indicating system instability).

Appendix D. Critical Analysis of Findings

D.1. Limitations

While our 400-session dataset provides substantial statistical power, several limitations warrant discussion:-

1. Results based on specific coding tasks may not generalize to all development contexts.
2. The complete security elimination in FF strategies suggests potential measurement artifacts or genuine extreme behaviors requiring further investigation.
3. Results obtained using GPT-4 may differ across other LLM architectures. We will expand the scope in a future study based on impending grants to support infrastructure costs.
4. FF strategy’s complete security elimination suggests measurement artifacts.

Appendix E. Future Research Directions

Our dynamical systems framework opens several promising research avenues including theoretical extensions to higher-dimensional objective spaces ($n > 3$) with eigenvalue degeneracy analysis, non-linear dynamics capturing saddle points and chaotic attractors, and stochastic control theory for optimal strategy adaptation. Algorithmic developments encompass real-time strategy switching based on eigenvalue drift monitoring, multi-agent extensions for collaborative LLM systems, and robust optimization with uncertainty quantification in drift and diffusion parameters. Broader applications include safety-critical systems with formal guarantees, human-AI collaboration modeling feedback as stochastic forcing terms, and multimodal integration analyzing text-vision-audio trade-offs. The intersection of dynamical systems theory with modern AI systems represents a rich domain for continued theoretical and practical advances, with immediate applications to next-generation LLM architectures and long-term implications for general AI system design.